MILITARY SPECIFICATION SHEET

FLANGES, FOUR BOLT SQUARE, BUTT WELD FOR HYDRAULIC SYSTEMS

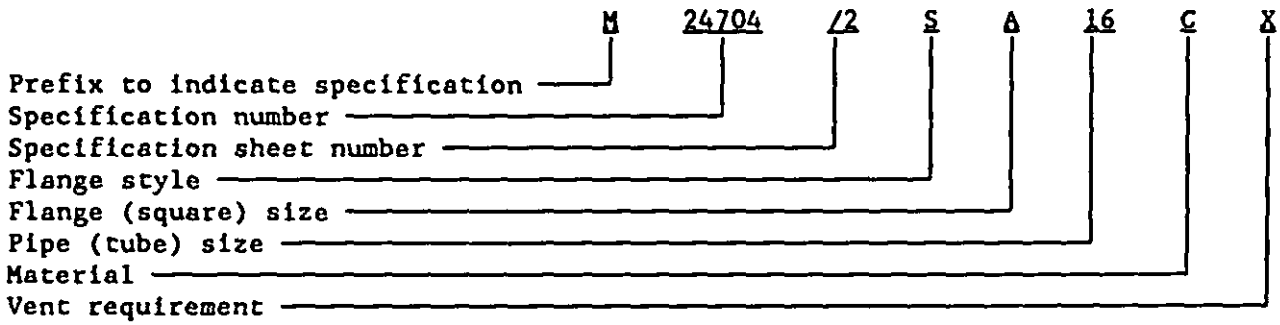
This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the flange described herein shall consist of this specification and the latest issue of MIL-F-24704(SH).

SCOPE

Scope. This specification sheet covers the requirements for four bolt square butt weld flanges for use in 3000 pounds per square inch (lb/in^2) hydraulic systems.

Classification. Butt weld flanges shall be designated by standard part numbers as identified below:



AMSC N/A FSC 4810

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Flange style. The flange style designation letter indicates whether a straight or angle flange and whether for tubing or nominal pipe size (nps) piping as indicated below:

S = Straight flange for nps piping (see figure 1)
L = Right angle flange for nps piping (see figure 2)
T = Straight flange for outside diameter (od) tubing (see figure 3)
R = Right angle flange for od tubing (see figure 4)

Flange (square) size. The square size of the flange shall be designated by a letter in accordance with the following list (see figures 1, 2, 3 and 4 for listing of compatible flange and pipe (tube) sizes):

A = 1.50 inches square
B = 1.75
C = 2.00
D = 2.25
E = 2.75
F = 3.50
G = 4.25

Pipe (tube) size. The pipe or tubing size shall be designated as a two digit number in accordance with the following list:

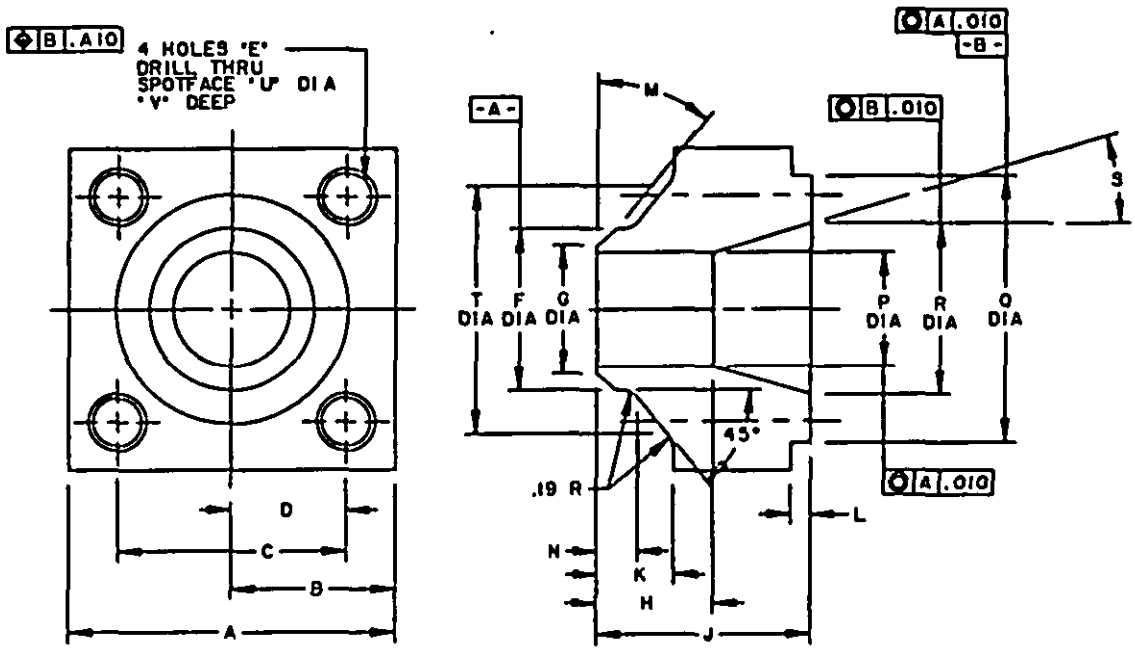
Size	Pipe (nps)	Tube (od)
01	1/8 (0.405 od)	---
02	1/4 (.540 od)	0.250
03	3/8 (.675 od)	.375
04	1/2 (.840 od)	.500
06	3/4 (1.050 od)	.750
08	1 (1.315 od)	1.000
10	1-1/4 (1.660 od)	1.250
12	1-1/2 (1.900 od)	1.500
16	2 (2.375 od)	2.000

Material. The flange material shall be designated by a letter as indicated below:

C = Corrosion resistant steel (CRES), class 304 or 316 in accordance with ASTM A 276, ASTM A 473 or QQ-S-763
L = Corrosion resistant steel (CRES), class 304L or 316L in accordance with ASTM A 276, ASTM A 473 or QQ-S-763
N = Nickel copper alloy in accordance with QQ-N-281, class A or ASTM B 127 (UNS N04400 only)

Vent requirement. The vent requirement shall be designated by a letter X, Y or Z as indicated on figures 5 and 6. For flanges without a vent the letter W shall be used. Machining for the vent port shall be as shown on the applicable figure.

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ALL TOLERANCES SHALL BE +/- 0.030

PART NUMBER ---- M 24704/2 S## ** xx
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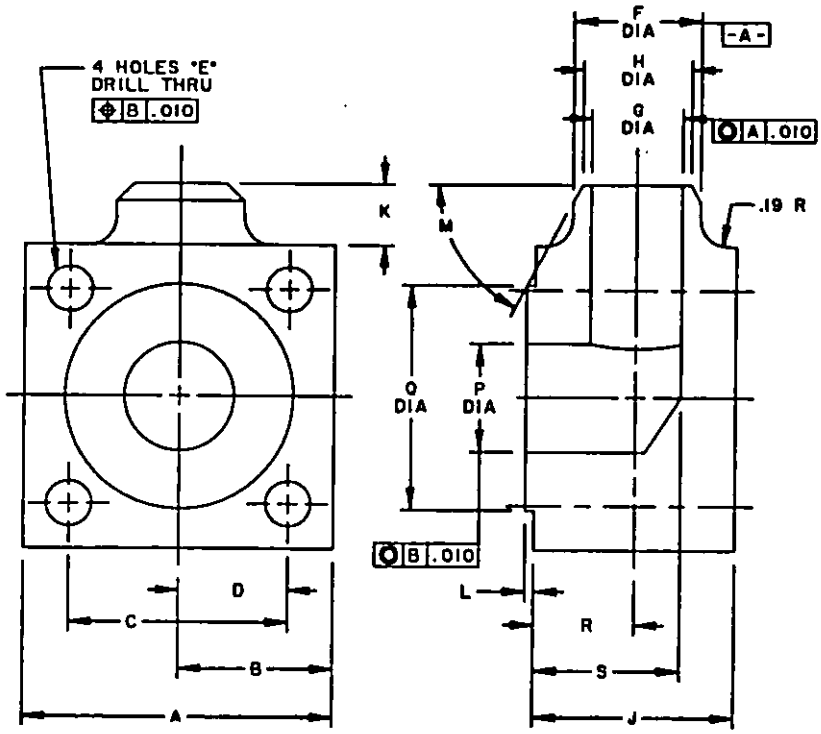
##	**	FLANGE SIZE (INCH)	PIPE SIZE N.P.S.	A SQ	B	C SQ	D	E	F	G +.010 -.010	H +.030 -.000	J	K
F	10	3.500	1.250	3.500	1.750	2.500	1.250	0.531	1.660	1.301	0.375	2.125	0.906
F	12	3.500	1.500	3.500	1.750	2.500	1.250	0.531	1.900	1.520	---	2.125	0.906
G	16	4.250	2.000	4.250	2.125	3.062	1.531	0.656	2.375	1.698	0.625	2.594	0.906

##	**	FLANGE SIZE (INCH)	PIPE SIZE N.P.S.	L +.002 -.000	M +5deg -0deg	N	P +.010 -.000	Q +.000 -.002	R +.000 -.015	S DEG	T	U	V
F	10	3.500	1.250	0.125	40	0.375	1.281	2.124	1.500	---	1.875	0.938	0.031
F	12	3.500	1.500	0.125	40	0.375	1.500	2.124	---	---	2.375	0.938	0.031
G	16	4.250	2.000	0.124	40	0.375	1.688	2.374	1.750	---	3.188	1.062	0.500

SH 13203057

FIGURE 1. Straight flange - pipe.

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PART NUMBER ---- M 24704/2 L# ** xx
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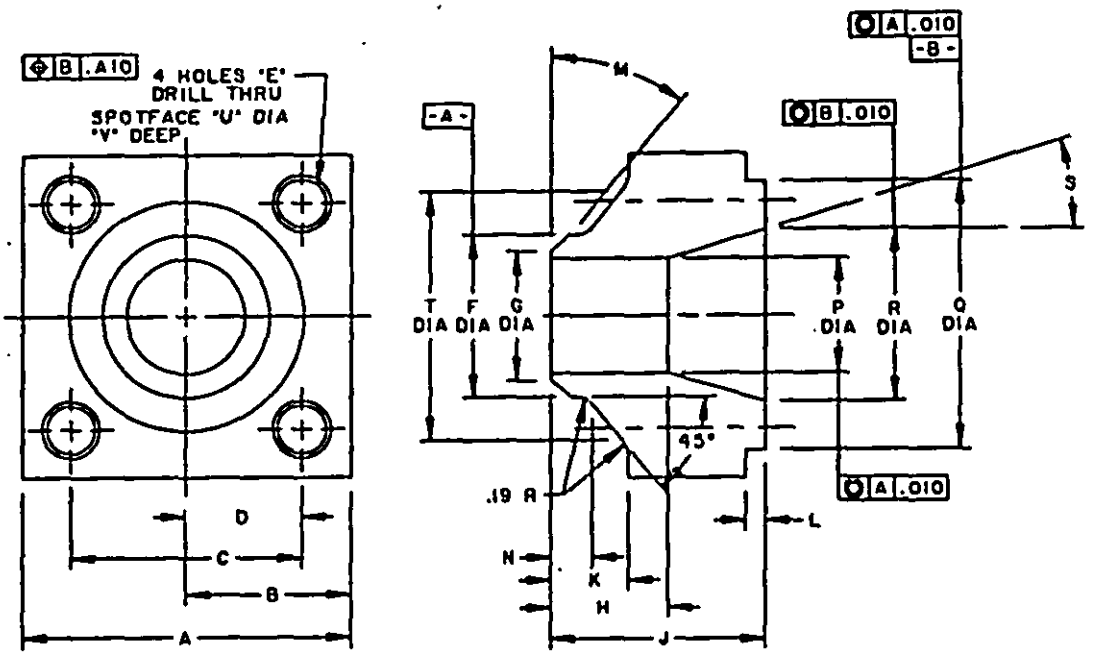
#	**	FLANGE SIZE (INCH)	PIPE SIZE N.P.S.	A SQ	B	C SQ	D	E	F	G +.010 -.010	H +.010 -.010	J	K
F	10	3.500	1.250	3.500	1.750	2.500	1.250	0.531	1.660	1.281	1.301	2.625	0.875
F	12	3.500	1.500	3.500	1.750	2.500	1.250	0.531	1.900	1.500	1.520	2.625	0.875
G	16	4.250	2.000	4.250	2.125	3.062	1.531	0.656	2.375	1.688	1.698	3.500	1.000

#	**	FLANGE SIZE (INCH)	PIPE SIZE N.P.S.	L +.002 -.000	M +5deg -0deg	P +.000 -.015	Q +.000 -.002	R	S
F	10	3.500	1.250	0.125	40	1.500	2.124	1.312	2.000
F	12	3.500	1.500	0.125	40	1.500	2.124	1.312	2.094
G	16	4.250	2.000	0.125	40	1.750	2.374	1.750	2.469

SH 13203058

FIGURE 2. Right angle flange - pipe.

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PART NUMBER ---- M 24704/2 T# ** xx
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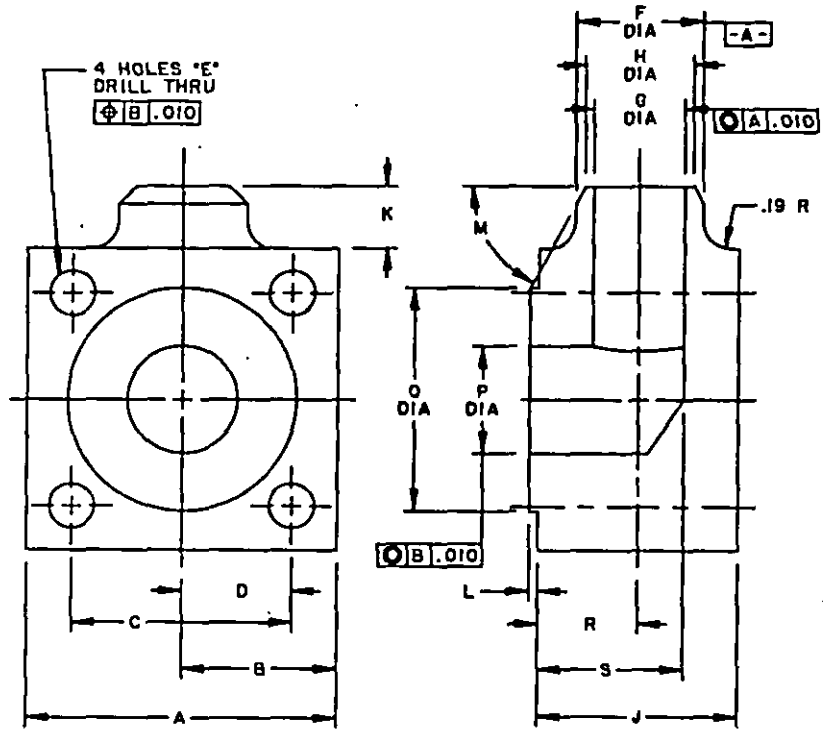
#	**	FLANGE SIZE (INCH)	TUBE SIZE	A SQ	B	C SQ	D	E	F	G +.010 -.010	H +.030 -.000	J	K
F	10	3.500	1.250	3.500	1.750	2.500	1.250	0.531	1.250	0.748	---	2.125	0.906
F	12	3.500	1.500	3.500	1.750	2.500	1.250	0.531	1.500	0.872	---	2.125	0.906
G	16	4.250	2.000	4.250	2.125	3.062	1.531	0.906	2.000	1.185	---	2.594	0.906

#	**	FLANGE SIZE (INCH)	TUBE SIZE	L +.002 -.000	M +5deg -0deg	N	P +.010 -.000	Q +.000 -.002	R +.000 -.015	S DEG	T	U	V
F	10	3.500	1.250	0.125	40	0.375	0.688	2.124	---	---	1.875	0.938	0.031
F	12	3.500	1.500	0.125	40	0.375	0.812	2.124	---	---	2.188	0.938	0.031
G	16	4.250	2.000	0.125	40	0.375	1.500	2.374	---	---	3.062	1.062	0.500

SH 13203059

FIGURE 3. Straight flange - tube.

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PART NUMBER ---- M 24704/2 R# ** xx
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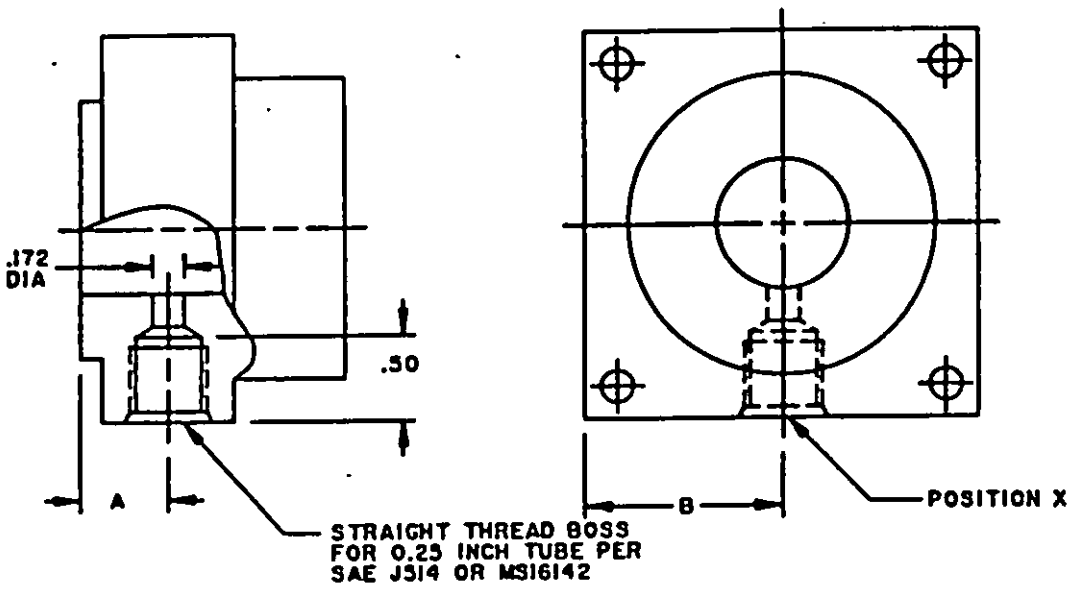
#	**	FLANGE SIZE (INCH)	TUBE SIZE	A SQ	B	C SQ	D	E	F	G +.010 -.000	H +.010 -.010	J	K +.015 -.000
F	10	3.500	1.250	3.500	1.750	2.500	1.250	0.531	1.250	0.688	0.748	2.625	0.875
F	12	3.500	1.500	3.500	1.750	2.500	1.250	0.531	1.500	0.812	0.872	2.625	0.875
G	16	4.250	2.000	4.250	2.125	3.062	1.531	0.656	2.000	1.125	1.185	3.500	1.000

#	**	FLANGE SIZE (INCH)	TUBE SIZE	L +.002 -.000	M +5deg -0deg	P +.000 -.015	Q +.000 -.002	R	S
F	10	3.500	1.250	0.125	40	0.688	2.124	1.312	1.656
F	12	3.500	1.500	0.125	40	0.812	2.124	1.312	1.718
G	16	4.250	2.000	0.125	40	1.125	2.374	1.750	2.312

SH 13203060

FIGURE 4. Right angle flange - tube.

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THE VENT PORT CAN BE LOCATED AT POSITION X FOR THE FOLLOWING FLANGES. INSERT THE LETTER X AT THE APPROPRIATE LOCATION IN THE FLANGE PART NUMBER. FOR FLANGES WITHOUT VENT REQUIREMENT, INSERT THE LETTER W.

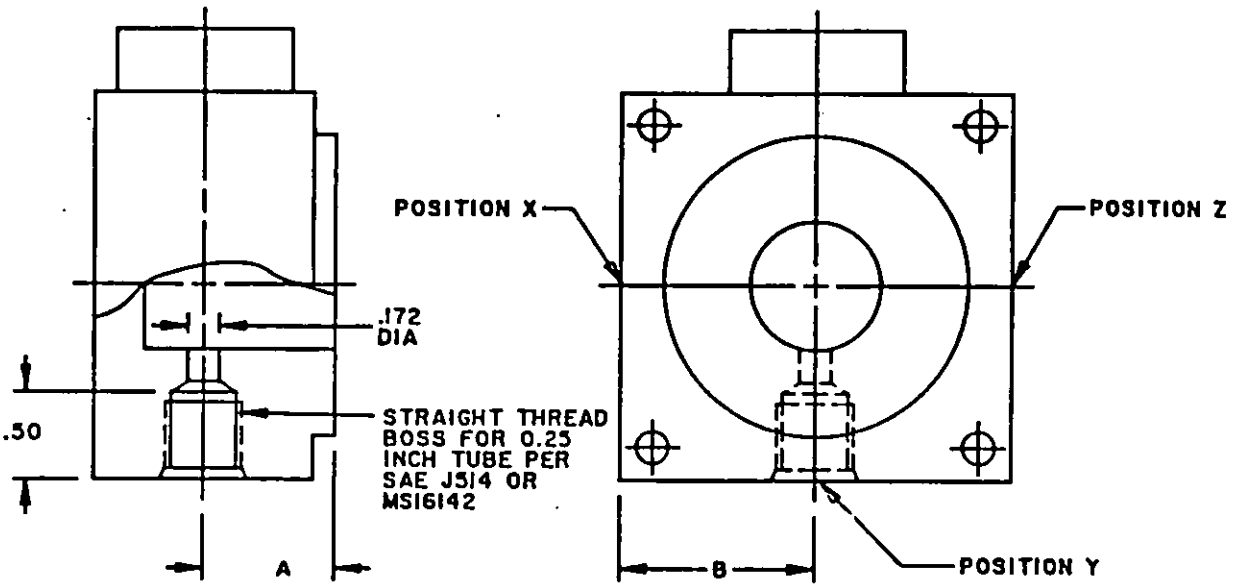
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ALL TOLERANCES SHALL BE +/- 0.030

# **	FLANGE SIZE (INCH)	PIPE OR TUBE	A	B
F 10	3.500	1.250	0.734	1.750
F 12	3.500	1.500	0.734	1.750
G 16	4.250	2.000	0.906	2.125

SH 13203061

FIGURE 5. Straight flange vent location.

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THE VENT PORT CAN BE LOCATED AT POSITION X,Y, OR Z FOR THE FOLLOWING FLANGES. INSERT THE LETTER X,Y, OR Z AT THE APPROPRIATE LOCATION IN THE FLANGE PART NUMBER. FOR FLANGES WITHOUT VENT REQUIREMENT, INSERT THE LETTER W.

UNLESS OTHERWISE SPECIFIED
ALL TOLERANCES SHALL BE +/- 0.030

# **	FLANGE SIZE (INCH)	PIPE OR TUBE	A	B
F 10	3.500	1.250	1.438	1.750
F 12	3.500	1.500	1.594	1.750
G 16	4.250	2.000	1.875	2.125

SH 13203062

FIGURE 6. Right angle flange vent location.

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